

Charting AI urbanism: conceptual sources and spatial implications of urban artificial intelligence

Otello Palmini¹  · Federico Cugurullo² 

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Abstract

The aim of this paper is to tease out some of the key issues concerning the relationship between AI and urbanism. This relationship, which is presented in the academic literature as a new driving force of contemporary urbanism, will be investigated through an interdisciplinary approach that places urban studies and philosophy of technology in dialogue. Thus, the analysis will not focus on the technological development of artificial intelligence systems but on how their application can affect urbanistic thinking and vice versa. The chart that is produced by this method is based on two fundamental axes: time and space. AI urbanism will then be inquired first through key turning points in the history of the relationship between technology and the city (modern urbanism, cybernetics and the smart city paradigm). Secondly, the spatial implications of urban AI will be investigated from the point of view of the concrete applications of this technology to the city (Robots, AVs, Software agents) and their impact on the relationships between different urban actors. Ultimately, this work aims to offer a conceptual tool for understanding some decisive implications of the relationship between AI and urbanism, such as the connection between quantitative and qualitative approaches, the implications related to autonomous technology, the economic-political background of AI urbanism, the material urban impact of AI, and the relationship between AI and other urban intelligences. Understanding these implications will be valuable for future research on AI urbanism oriented toward transforming simple technological development into sustainable urban innovations.

Keywords AI urbanism · Artificial intelligence · Cybernetics · Smart city · Urban AI

1 Introduction

Artificial intelligence (AI) is nowadays a common buzzword in both academic and public discourses. AI certainly belongs to the radical technologies that are capable of fundamentally altering the material composition and our interpretation of everyday life [45]. The aim of this paper is not to contribute to the debate concerning the technological development of AI, but rather to increase the comprehension of the stakes concerning the relationship between this technology and urbanity. In fact, this connection is deemed by many scholars to be the next driver in urban development, sparkling at the same time hope and harsh criticism [61, 6, 26, 62, 110]. By the term AI we generally mean a category of artefacts capable of acquiring information from the surrounding environment and making rational decisions autonomously and even in situations of uncertainty [17, 25]. The tools in use today are narrow AIs—that is, systems capable of dealing with specific domains and problems for which they are trained. The urban impact that these technologies can have is extremely

✉ Otello Palmini, plmtll@unife.it; Federico Cugurullo, CUGURULF@tcd.ie | ¹Department of Architecture, University of Ferrara, Ferrara, Italy. ²Department of Geography, School of Natural Sciences, Trinity College Dublin, University of Dublin, Dublin, Ireland.

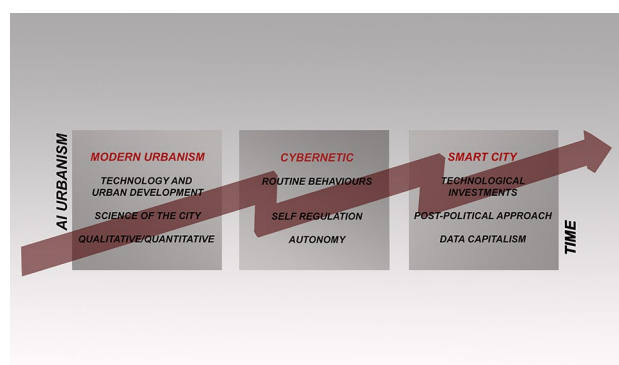


significant. AVs, robots and software agents are becoming widespread forms of agency in urban fabrics. The functioning of cities may be changed by the use of these tools in various domains: economy (making processes more efficient and increasing productivity), society (community health monitoring and autonomous mentoring systems), environment (optimization of material and immaterial flows), governance (decision support systems, surveillance and cybersecurity) [11, 46, 112]. However, these great opportunities imply that equally significant challenges need to be addressed not only from a technological perspective but also by developing a cultural and project-based understanding of the relationship between AI and the city. Because of this reason charting AI urbanism here does not mean enumerating the empirical *loci* (cities, regions, states) where this relationship is taking place, but mapping some research trends and their conceptual sources. Charting this relationship means exploring how certain structural elements of AI may affect the way we understand the city and, also, considering how a certain interpretation of urbanity may affect the manners in which this technology is deployed. This chart is meant to be a critical conceptual tool applicable to a relationship, that between AI and the city, which contemporary debates often understand as natural, fluid, and frictionless, by highlighting its depth and problematic relevance.

In the remainder of the paper, the coordinates of this map will be time and space. Time understood as a set of theoretical models, afferent to both urban studies and philosophy of technology, that influence the relationship between AI and urbanism. The spatial coordinate, on the other hand, will concern as much an investigation of the urban materialization of AI as its influence on all those relationships among humans, the built environment and nature that constitute the flesh of the urban fabric. Thus, from a temporal point of view the aim will be to comprehend how certain historically determined conceptual sources exert an influence on the relationship between AI and urbanism. First, this relationship will be interpreted through some key features of modern urbanism. Particular attention will be devoted to the relationship between scientific knowledge and the city by pointing out continuities and discontinuities between the science of the city as conceived by modern urban planners and the one structured around Big Data and AI. A second focus will be on the role of cybernetics as a conceptual source of AI urbanism. The cybernetic theoretical framework, developed since the middle of the last century, will be interpreted as a valuable conceptual source that can clarify some aspects of the notion of autonomy that imbues the relationship between AI and cities. The section on time will complete the chart by considering the most recent model of AI urbanism, namely the smart city paradigm. Here the focus will be on the economic dynamics that have marked the rise of this paradigm and how the massive deployment of AI in the urban context can learn some lessons from the academic debate that has developed over the past fifteen years on the relationship between digital technology and urban issues (Fig. 1).

This first section devoted to providing a historical background of the relationship between AI and urbanism will be followed by one dedicated to understanding its spatial impacts. First, some characteristics of AI that make the very use of this technology spatially determined will be highlighted by connecting them to its urban incarnations, paying attention to how these can alter the contemporary urban experience. Finally, the second section on space will consider how AI is placed in relation to other urban intelligences (human and natural). This section will suggest a theoretical paradigm shift capable of overcoming a reductionist view of the relationship between urban intelligences toward a more sustainable approach. The purpose of this paper is not to produce a literature review on AI urbanism. The aim is to develop a dialogue between a number of disciplines (Philosophy of Technology, Urban Planning, Sociology, Urban Geography) to highlight implicit assumptions and possible implications of the relationship between AI and cities. In the first section, the one devoted to the time axis, our insights highlight some of the main recognized theoretical sources of AI urbanism and analyse them in an interdisciplinary way to bring out both a critique and a development path for the paradigm in

Fig. 1 Temporal coordinate: Modern Urbanism, Cybernetics and Smart City as sources of AI Urbanism. Source: Authors' original



question. In the section devoted to space, on the other hand, we decided to focus on two concepts that we consider decisive for understanding the relationship between AI and Urbanism: the materiality of urban artificial intelligence and the very concept of intelligence that is implicated in this debate. We believe that combining a critical perspective with a historical one can contribute to a better comprehension of the conceptual genealogy and thus the lines of development that characterize AI urbanism.

1.1 Towards a science of the city: from modern to AI urbanism

It is well known that urbanization processes are deeply connected to technological development [82]. The ability to construct buildings, to develop infrastructure, to seize space from the natural world and to annex it to the urban realm are factors that can only be understood by paying close attention to the technological development of societies [81]. However, in Western modernity, we have been witnessing a shift in the quantity and quality of the relationship between technology and society. The turning point that marked this decisive shift in Western cultural history and in the history of urban theory is represented by the work of Francis Bacon (1561–1626) [111], who can be considered as a forerunner of the current understanding of the relationship between technology and society that today affects our attitude towards AI in cities [26]. Indeed, in his urban and social utopia called the New Atlantis, Bacon portrays science and technology as the engine driving the improvement of society. Here technology is no longer just an important factor in the urbanization process: it becomes society's governing principle and the dominant force capable of promoting human development. In his philosophy, Bacon replaces Plato's philosopher kings with a class of scientists capable of governing the urban realm and keeping it healthy, in virtue of their technical and scientific expertise. As Coeckelbergh [19] remarks, the idea that today AI enables an objective and impartial process of decision-making that should be placed at the centre of society is nothing more than a technologically advanced version of this century-old utopian thought.

This Baconian seed sprouts powerfully in 19th-century urban planning. More specifically, modern theorists understood the city as a process of turning chaos into order, by means of scientific inquiry and technological innovation [26]. Moreover, the spatiality of the modern city was then deeply informed by the scientific and technological disruptions triggered by the industrial revolution [8]. Already at the origin of modern urban planning, for example in the work of Idefonso Cerdà (1815–1876), there is an aspiration for an articulate scientific approach in the urban realm [15]. Statistical surveys, topographical representations and written descriptions form the basis of this new mode of investigation [115]. In this context, the need is to develop an instrument of analysis and a language as objective as possible to describe and thus intervene in the urban fabric. This desire for objectivity dialogues with scientific methods and with the technological innovations of the nineteenth century, in the attempt to establish a true science of the city. In this century characterized by the development of analytical and historical sciences, these two dimensions contribute to structuring a new objectivity for urban analysis.

According to Duarte and Alvarez [30], this nineteenth century logic is also at work in the era of Big Data and artificial intelligence. In fact, both the ability to store data through IoT systems and the power to analyse it through AI mechanisms would allow us to develop a new science of the city [63, 68]. This would be a digital and algorithmic urban science with different methods and analytical tools than Cerdà's but one with a similar intention. In this contemporary model, cities become the optimal context for using artificial intelligences due to the large amount of data they express, and artificial intelligences become decisive because they provide solutions to articulated problems that are too extensive for other types of analysis. Just as statistics had been able to contribute to the objectivity of modern analysis, today the ability of AI to generate knowledge becomes decisive for a new representation or construction of urban reality [19].

The struggle for objectivity that characterised the nineteenth century was certainly driven, as we have seen previously, by the application of new scientific methods of analysis and technological innovations. However, quantitative analysis—inherent to statistics and topography—was then supplemented with the qualitative character of iconic written descriptions. The power and evocative capacity of Engels' descriptions or the novels by Zola and Dickens are certainly decisive for the representation of the industrial city by bringing out a missing aspect in mere statistical data. Zola's descriptions of the atmospheres created by the new organisation of space and the relationships established in it, for example, have contributed to the understanding of Haussmann's Paris. In Dickens' pages the counter-history of the industrial city takes shape bringing to light the misery, exploitation and insalubrity hidden under the veil of glittering progress. Similarly, from the descriptions of Engels, we have a glimpse of the living conditions of the urban proletariat, a new class destined to enter history in a disruptive way.

It is no coincidence, then, that currently in the face of the vast amount of quantitative analysis that artificial intelligences are capable of conducting, one key issue is how to strike a balance between this quantitative approach and qualitative aspects [30]. As Coeckelbergh [19] claims, when it comes to political judgements (which are intrinsic to urban planning interventions) scientific knowledge is necessary but certainly not sufficient. Hence, any new AI-based science of the city should be constituted by an interdisciplinarity capable of balancing the quantitative parts with qualitative analysis. However, this dialogue must not only be conducted *ex post*, after quantitative analyses. It is necessary to include the qualitative dimension provided by the social sciences and humanities in the process of codification of artificial intelligences. From an ethical point of view, there are two interesting and complementary approaches: *ethics by design* and *ethics in design* [38]. The former consists of consciously embedding ethical principles within autonomous decision-making mechanisms; the latter concerns cultural training that includes ethics for technicians working on the technological design of AI.

The relationship between AI and the social sciences also needs to be deepened. Indeed, the latter should not only work on the implications and consequences of the application of the former. The social sciences must be brought into the design and governance of artificial intelligences to formulate conceptual models and to reflect, already at the coding stage, on possible medium-term issues [20]. The path to a new science of the city, capable of balancing quantitative and qualitative approaches, also involves transdisciplinarity. Not only interdisciplinarity, that is the connection between various disciplines, but also transdisciplinarity as a two-way movement between society and specialisms [63]. The democratic involvement of society in structuring such new science will be pivotal not only as a feedback mechanism but also as a vector for identifying problems and designing solutions. This first act of charting AI urbanism via modern urbanism should stimulate us to achieve a hitherto missing balance between quantitative and qualitative urban aspects, between the algorithmic ability to give solutions and the social capacity to understand problems and shape the possibilities of solutions. Moreover, this historical perspective should warn us against a deterministic drift in the science of the city and spur us to think about a hybrid and truly multidisciplinary approach to urban development.

1.2 The seductiveness of mechanical autonomy: cybernetic sources of AI urbanism

The challenge of balancing a quantitative approach with a qualitative one is present, in a different form, at another fundamental step of the modern history of technology. In the middle of the twentieth century, starting with the seminal work by Bigelow, Rosenbleuth and Wiener in 1942, cybernetics took shape [54]. This discipline, based on the concepts of control, information, feedback and entropy, aims not only to explain the functioning of certain types of machines, but above all to become the common ground for interpreting the behaviour of both organisms and machines [109]. This new perspective is based on what Wiener calls *the second industrial revolution*. This new disruption does not merely involve the mechanical replacement of human physical force, as in the previous industrial revolution. What Wiener [109] refers to is an unprecedented technological innovation that makes it possible to mechanically replace functions that have traditionally been attributed solely to the human brain. The new machines described by Wiener can communicate with their environment through sensors and correct their behaviour in relation to a purpose, through feedback mechanisms. According to cyberneticians, this scheme would be able to explain teleologically oriented patterns of action (intended as purpose-driven actions) regardless of whether these are carried out by a machine or an organism. This is because the scheme is abstract and consists of concepts, such as *system*, *control*, *information*, *feedback*, *input* and *output*, that are independent of the material composition of the acting subject [48]. Following the reflections of Jonas [60], it is possible to claim that this characteristic allows cybernetics to present itself not only as a new theory for machines, but as a unified theory of reality.

Cybernetics was born and developed in a historical and geographical context in which stability became the main goal for society [4, 54]. This new discipline provides a model of automatic regulation that is particularly attractive from this perspective. However, Wiener himself had been critical of the social application of cybernetics. In fact, he believed that the great complexity of the problems and the scarcity of data would limit the practical use of cybernetics [109]. However, subsequent technological development, in terms of the ability to acquire data and the capacity to analyse it, led to the idea of a possible cybernetization of urban reality. A reference model is certainly Jay Forrester's *Urban Dynamics* [42], which, however, still suffers from a significant dissonance between an idealized technological representation and the reality of cities. Nevertheless, this theoretical-interpretative paradigm has remained the foundation of a viable connection between technology and the city. Moreover, its claims have become more realistic in the contemporary scenario marked by an increased computational power coupled with a huge capacity to acquire data [87]. Dwelling on the definition of urban artificial intelligences (UAI), it is possible to recognise the influence of cybernetic concepts on

the way AI is today implemented in the urban realm: “UAls are autonomous artefacts capable of gaining knowledge about the surrounding urban environment and making sense of acquired data, using it to act rationally according to pre-defined goals in complex situations and spaces in which some information might be missing and, above all, humans are not steering their actions” [26].

Thus, today UAls are machines capable of interacting with their environment and regulating their behaviour in relation to pre-defined goals, in line with the ideals theorized by cyberneticians in the past century. In fact, some scholars (urbanists in particular) have noted a direct connection between the application of AIs in cities and the history of cybernetics [4, 55, 75, 87]. Two elements emerge prominently from these studies. Firstly, the interpretation of the city as a set of purpose-driven actions in the cybernetic sense, routine behaviours, which can be replicated, predicted and analysed by UAls [4, 87]. Secondly, AI systems can be interpreted as mechanisms capable of coordinating and stabilising all agents present in the urban context under the same logic, which is a perspective that resonates with cybernetics as *the science of control* meant to harmonize predictable routine behaviours [75].

However, this perspective raises some critical issues. Firstly, it should be noted that routine behaviours are indeed a core part of what happens everyday in the city, but they are not everything. The urban fabric is also made up of contingencies, unpredictability and non-harmonisable relationships [71]. Focusing only on the former and underestimating the latter leads to a misunderstanding of urban reality that overshadows its potential for creativity and unpredictability [4, 91]. Furthermore, critical urbanists have pointed out that in most cases AIs operate on the basis of datasets that refer to past actions and choices, and this can lead to an autonomous perpetuation of bias and omissions [4]. The risk is to perpetuate discrimination and inequality, which is an already present urban issue that has been identified in the housing market [90] and predictive policing [92, 103].

These problems result from a flawed assumption concerning the very notion of autonomous technology. According to Winner [109], a technology is autonomous if it is not controlled by something external to it. For example, a technology is not autonomous if it is controlled by human will. Winner explicitly mentions the cybernetic perspective as an eminent framework for this conception of technology. Today this claim to autonomy returns in the age of artificial intelligences equipped with more computational and predictive power than in the 1970s [26, 79]. However, as Jonas [60] had already noted, from the very beginning cybernetic reflection suffered from a confusion between machines capable of serving a purpose and machines capable of having one. According to Jonas, cybernetic machines can fulfil the former function but not the latter: the impetus for the pursuit of a purpose never comes from a necessity intrinsic to the machine but always from an input external to it. This perspective has also been reiterated in philosophy of technology with regards to AI systems [36]. This kind of confusion about what purposes should be pursued also affects the use of AI in the urban fabric. Indeed, in the case of UAI, this line of reasoning risks overshadowing two things: on the one hand, the fact that a city does not have a single purpose to pursue but has several ones that emerge through discussion or conflict between urban actors [44]; on the other hand, the fact that a future in which UAls will be able to act in an unsupervised manner could be affected by the structural problems of this technology (partiality of data, bias and discriminations) or even lead towards the end of the city as something traditionally designed by humans for humans [26].

This second cybernetics section of our chart leads us to question the idea that the city can be interpreted as a homogeneous and discrete matter. This problematic idea conceives the city as an autonomously calculable entity, and this leads to the belief that algorithmic and autonomous decisions are devoid of partiality. This logic legitimates a utopia in which autonomous decision-making mechanisms can replace humans’ planning capacity. However, the analysis of the history of cybernetics can also be used to propose alternative models of the relationship between AI and the city. This discipline has had its own historical evolution marked by fractures and conceptual innovations that are still ongoing today [51]. Even if the reference to cybernetics in urban studies is often linked to the notions of control, stability and the absence of freedom, this is not a mandatory route. The search for a radical, open cybernetics that is attentive to the dynamic connection between different intelligences instead of their forced harmonisation is still open [22, 35, 57, 75, 94, 113].

1.3 Smart sources of AI urbanism

As we have seen in the first two sections of our charting process, the relationship between the city and technology has deep roots and it has gained new impetus from the modern age onwards [26, 115]. We have also observed how the emergence of cybernetics influenced this connection by generating projects aimed at the computational management of the urban context [55, 87]. However, if we refer to digital technology as we understand it today, the first reflections on its use in cities date back to the 1990s [110]. Mitchell [78] considered the new architectural and urbanistic opportunities

derived from the digital integration of urban space, the so-called *City of Bits*. Similarly, Maldonado in 1997 was thinking about a potential infrastructural role of digital technology in the urban context [70].

However only the rise of the smart city paradigm in the first decade of the twenty-first century marked the real turning point of the connection between digital technology and the city in terms of the amount of investments, projects and diffusion of ideas [13, 97]. IBM, one of the leading companies in the field, used to define a smart city on the basis of three criteria: *instrumented, interconnected, intelligent* [51]. While it is not easy to find an unambiguous definition of smart city, it is certainly true that smart urbanism is enabled by the use of ICT to tackle and supposedly overcome a number of urban issues such as mobility, waste and energy management, governance, economic growth, participation in social life, and security [2, 21, 26]. The rise of this paradigm and the beginning of the promotions of smart solutions by some companies has emerged in the post-crisis economic situation of the late 2000s in a context in which public administrations were struggling to invest. In this context, private companies proposed their technological tools to policy makers as solutions to the problems of waste of resources and various systematic inefficiencies within cities [58]. Moreover, this process was accelerated by the necessity of cities to brand themselves as *smart* and at the cutting edge of technological innovation in order to obtain favourable loan interest rates [13] and to increase their chances of accessing European or government funds [104]. In this rationale of international competition between cities, the only way to go with the flow of smart urbanisation seemed to resort to private capital.

According to the narrative created by the big companies that dominated the smart city market in the first decade of the century (IBM, Cisco, Siemens, for example), the urban fabric should have been understood as a system of systems that had to be made as efficient as possible through the use of ICT [97, 62]. The implementation of digital technologies was presented as a technical development, without political or economic implications, towards a new era of urban liveability [87]. However, the other side of the coin of this process is that the implementation of essential digital infrastructure is managed by private companies and thus legitimately driven by a desire for profit [97, 86]. Furthermore, this approach to the relationship between the city and technology has been criticised by various scholars for a reductionist conception of the city [16, 44, 91], for issues related to a lack of consideration of the social demands expressed by citizens [102, 104] and for the risks associated with surveillance and the possession of data [43]. In fact, this post-political approach to urban development, unable to grasp changes in political and economic power relations, has paved the way for a development of the smart paradigm centred on the actions of private companies and on neo-liberal economics [3, 97]. This logic led to a situation where vital urban infrastructure was owned by private companies and where private platforms proposed themselves as market alternatives to fundamental issues such as housing [96] and mobility [13]. In all these cases there was an extraction of value from cities on the part of these companies under the banner of a politically and economically neutral technological development [44, 73, 102]. Many of these issues are related to the possession and use of data produced in urban contexts [43]. In fact, when the digital infrastructures and platforms operating in the city are privatized, the collection and use of data are inevitably available to private companies and not to communities [93]. This theme cuts across the literature on smart urbanism and is key to understanding the contemporary relationship between AI and urbanism.

This ability to generate massive amounts of data makes the city a decisive component in the future evolution of AI. In this technical and economic context, the abundance of data and thus cities become crucial for companies both to create value in this phase of capitalism [114] and to boost the potential for AI development [66]. The impact of AI on the economic dynamics of capitalism is still uncertain. However, the risks of its deployment in the context of existing neo-liberal economic discourse are already visible [76, 65]: monopolistic tendencies, price discrimination, increasing pre-existing social inequality [64], but also privacy issues, political manipulation, bias and discrimination, (self-) disciplining by data [19]. Under these conditions it seems difficult to ignore the problems created by the smart city paradigm and then replicated in AI urbanism. Even outside the Western context, these issues do not lose importance. An interesting case study is Hong Kong which formalised its smart city programme in 2013 and in 2018 recognised AI as a key asset to invest in [56]. However, the projects related to these initiatives have proven to be fragmented and poorly connected to the social and environmental context mainly because of a lack of public regulation of private actors and the preponderance of a neoliberal approach [26].

On the basis of these findings, it is crucial to develop a model for analysing the impact and economic management of AI that is capable of taking into account the altered industrial environment, the new ways of producing value and the resulting different social structure. Indeed, these elements inform the production of contemporary urban space. A viable perspective is to expand the concept of digital commons in the sense of data commons and to make public investments to put computational power at the service of communities [107]. This approach refers to the common and distributed ownership of digital assets [9] that allows citizens to access and use data as a common good [12]. This perspective applies

to the notion of data justice, which pursues the avoidance of unfair outcomes in the use of data, the focus on the creation and accessibility of data sets, and the problems of privacy and rights violations associated with the use of data [53]. In a broader sense this approach is referring to an update of Levebvre's famous concept of the *Right to the City* [67], the idea being to involve citizens in the production of urban space through active and conscious use of digital infrastructure [28]. In addition to these processes, which have already been experimented in urban contexts [5, 49], investment in the education of professionals and decision-makers who are competent and aware is also pivotal.

Thus, from this section of the chart it is possible to understand that the future of the relationship between AI and the city passes through a rethinking of the economic and political context that has informed the smart city paradigm. The narrative of a necessary connection between technological development and social welfare must be replaced by a structural analysis capable of relating specific technological solutions to the economic, social and environmental contexts of application: stakeholder engagement and network, agility issues connected with AI agency, monopoly risk, lack of ethical frameworks for urban AIs, need for public regulation, the aspiration of an AI for social good [112].

1.4 Materializing urban AI

After investigating AI urbanism through time via modern urbanism, cybernetics and the smart paradigm the next two sections focus on space. The relationship between AI urbanism and space will be drawn by examining firstly its material embodiments and secondly its influence on the intangible relationships that innervate the urban fabric (Fig. 2).

Substantial technological transformations have always had a major impact on the organization of urban space [47, 82]. Cities' economies, the built environment, and urban social relations themselves have often been reshaped in relation to technological shifts [72]. This aspect was particularly evident in industrial modernity and its spatial unfolding in the first half of the last century [8]. Certainly, the digital revolution has triggered a disruptive change affecting multiple spheres of our experience [37]. However, especially in the passage from the 20th to the 21st century, the supposedly immaterial nature of information technologies seemed to undermine the material impact of this particular technological disruption. This has led to the interpretation of a new, light and fluid modernity that has had in Bauman [7] its most eminent theorist [26]. Despite their pervasiveness, ICTs did not have a significant material impact on cities at least until the beginning of the new century [50], thereby justifying some interpretations regarding the possible disappearance of the material centrality of the city in the age of interconnected economies and digital space [84, 85]. However, as early as 1996 in *City of Bits*, a landmark book for digital urban studies, a tension emerged between immateriality and materiality in the digital hybridization of the city [78]. While Mitchell highlighted a potential digital despatialisation, he was already thinking about how to interface physical and digital space by indicating a new and promising way forward for urban design.

This tension between the materiality and immateriality of digital technology also marks the contemporary AI debate: AI certainly constitutes a radical technology capable of pervasive insertion into our everyday experience [45]; moreover, it is possible to chart a material atlas of the fabrication and concrete implications of this technology [24]. Even though the allusion to intelligence enhances the perception of immateriality and lightness, the concept of *artificial* points to something manufactured, to a material assemblage and, ultimately, to something transformed through labour into an artefact. This tension also exists when it comes to UAI [11, 25]. In this context, recent empirical

Fig. 2 Spatial coordinate:
Implications of the materialization of urban AI and effects on urban relationships.
Source: Authors' original



studies highlight a materiality that is difficult to underestimate. An extensive body of literature on urban robots [77, 101], autonomous vehicles [29, 14, 27], and software agents [92] exposes specific material embodiments of this technology, strongly rejecting the perception of immateriality. These new artificially intelligent urban actors come in contact with our urban existences in different ways, thus raising the issue of regulating interactions that can safeguard their functioning but also our liveability.

Moreover, it is necessary to consider another spatial characteristic of AI. Philosopher Luciano Floridi [37, 40, 41] argues that the increasing efficiency of AI is due to a specific spatial action. In fact, the recent successes of AI systems not only depend on an increased technical performance that enables a better understanding of the world, but also on the actions that we undertake to make reality more intelligible to these systems. Floridi refers to this feature as *enveloping*. In this kind of reshaped spaces an AI is facilitated to gain information and act rationally and effectively. This process of AI spatialization establishes a necessary connection between AI functioning and spatial modification, thereby contributing to a material interpretation of this technology.

When it comes to UAI, the modes of this spatialization are the decisive background for a fundamental difference between *automated* and *autonomous* technologies. Automated technologies are enveloped in a space entirely dedicated to their action: a simple space not shared with humans or with anything that might perturb machines' functionality; in contrast, autonomous technologies act in complex spaces, shared with other forms of intelligences [26]. The envelope of automated technologies can be entirely designed according to the functional requirements of the machine. The designer has to focus only on increasing the efficiency of the system through a topography capable of maximizing its potential. The case of autonomous technology is quite different: technological envelopment must deal with other forms of independent agency (humans in particular). These actors have their own and specific purposes, which are not limited to maximizing the efficiency of the technical system. Take for example the case of an AV supposed to operate in a common urban street. Certainly, the designer will adapt the space to the requirements of the AV. However, this is not the only form of agency occurring in an urban street: pedestrians, cyclists, shopkeepers and plants are also decisive elements of this environment. These are just some of the potential actors whose agency is obviously unrelated to the efficiency of the action of AVs. Thus, the design of the spaces in which autonomous technologies are embedded cannot be limited to technological efficiency; it must consider a significant amount of variables and make precise design choices to address the problems of coordination among multiple urban actors [89].

These are common issues in urban experiments where small parts of cities, or entire urban areas, are used as test-beds for the implementation of UAI. It has been rightly argued that the problems to be addressed in these spaces are not only technological, since they also concern the physical composition of space, the geography of relationships between actors, governance, legal aspects related to the use of new technologies, and the ethical issues due to the interplay between different forms of agency [108]. Recent technical breakthroughs in AI have made it smoother for these systems to operate out of confined and restricted spaces [95], by facilitating the transition from automated to autonomous UAI. The increasing use of robots in the urban context is triggering a new direction of research on Robot City Interaction (RCI) that considers the possibilities of interaction between different types of robots and certain dimensions of the urban experience [101]; human–robot interaction (HRI) in public spaces, taking into account the trust that these agents should inspire in citizens, the integration of robots into the actual design of public spaces, and finally the equal distribution of the value generated by these agents [77, 99]. Research on AVs has also been investigating the relationship between human agency and AI systems by focusing on the possible uses arising from the encounter of these two forms of agency. For example, this is the case of how the use of the new space within AVs may affect the relationship between labour and non-labour time [74]. Finally, the increasing use of drones has also been investigated in urban studies with a focus on the risks of widespread surveillance and governmentality of common spaces [59].

Based on the first spatial section of the chart, we can draw the conclusion that AI, like many other disruptive technologies, has an extensive material component. When it comes to UAI, this materiality is particularly evident in enveloping processes and in the multitude of embodiments through which AI systems acquire their own agency in the urban fabric. This undeniable impact on urban space means that we must consider UAI not simply from a technological perspective. Designing spaces just for the sake of increasing the efficiency of technological systems is already difficult in the case of automated devices, and it is certainly harmful when it comes to autonomous ones. We argue that teasing out the materiality of UAI and its possible implications for urban spaces is key to properly set up the debate on its integration into everyday urban experiences. Conversely, treating UAI as an immaterial technology can only lead to a socially and environmentally detrimental design.

1.5 The (urban) geography of relational intelligences

The spatial impact of AI on the urban fabric does not only involve material spaces. As we have discussed in the previous section, it is certainly relevant to consider the material impact of this technology. However, we must also inquire how UAI may affect urban space from a *relational* perspective. That is, the quality and number of intangible relationships between urban agents that contribute to structuring the flows of knowledge and practices that characterize a specific urban realm. In other words, the impact of AI is perceptible not only on the *ville*—the physical components of the city—but also on the *cit  *, that is the way inhabitants experience urban space [91]. AI is often understood as a form of intelligence that rationalizes the interaction between different agents in the urban context. Indeed, in different spheres of urban life, which we can consider as socio-technical assemblages of interactions, AI interventions are intended as ordering and efficiency-bringing. This is because AI skills are considered to be superior to human ones and capable, therefore, of establishing greater rational efficiency.

This relational preponderance through which AI is interpreted is based on a precise theoretical framework. Indeed, building on the notion of *transhumanism* [10]—the possibility of going beyond human limits and abilities through technology—this approach to the city has been called *transurbanism* [26]. This approach promotes AI technology as a solution framework for urban questions, while overshadowing other intelligences (human and non-human) acting in the urban context. However, this perspective is based on a moot assumption: namely, that artificial and human intelligences are comparable and that, for this reason, one can quantitatively or qualitatively surpass the other. It has been pointed out that there is limited theorization on the concept of intelligence involved in the notion of AI, especially in urban studies [69]. The supposed intelligence attributed to AI is based on a cognitivist interpretation that is questioned within the psychological debates themselves as a model for interpreting human intelligence [69]. In addition, sociologists and philosophers of technology have been strongly questioning, even from different standpoints, the notion of intelligence as an appropriate metaphor to understand the powerful output-giving skills inherent in AIs [32, 40]. According to Floridi [40], for example, AI's recent successes are not based on the growth of intelligence, but on a divorce between the ability to perform complicated tasks and the need to be intelligent to do so. AIs then should be understood as new effective forms of agency, not intelligences. Also, according to communication sociologist Elena Esposito [32], the recent triumphs of AI are not motivated by the pursuit of intelligence. Indeed, programmers in their daily work neither aim to mimic human intelligence nor to build machines that can surpass it, but they rather target specific problem-solving techniques. Through this perspective, AIs have become important communicative partners for humans even in the absence of any kind of intelligence. In both cases, regardless of the differences in approach, AI is considered not as an intelligence but as an entity (agent or communicative partner) with which to relate but unable to surpass or be comparable with human intelligence.

However, in the urban context the technocentric view has led to considering AI as the only or predominant intelligence in the urban future [69]. This perspective grounds urban development on the quality and quantity of technological advancement [26]. The more technology is pervasive and capable of effective agency, the more urban development will follow a desirable and rational direction. AI urbanism seems to suffer from a misrecognition of the various forms of intelligence already present (the series of already established interactions among humans, the natural and the built environment) in urban systems, by rarely taking into account the ways in which UAIs might relate to these in a nonreductive way [11]. A very promising perspective to overcome this technocentric approach is certainly to consider the city as a process whose rhythm is marked by the relations between different forms of agency [33, 34]. This perspective instead of hierarchically reducing urban actors invites the tracking of social-technical-natural assemblages through an operation in continuity with the mapping proposed by the present paper. Assemblage theory focuses on the variagated composition of urban entities paying attention to the practices and networks in which they are situated [34]. Following this path would be as useful from both a theoretical and practical perspective [33]. Indeed, from a theoretical point of view, technological reductionism would be avoided, and from a planning perspective projects could be based on a comprehensive knowledge of the relationships in which they should be embedded.

One example of this misrecognition and technocentric approach concerns urban sustainability. The reflection on this topic has very different conceptual sources in comparison to the technocentric perspective that informs the discourse on smartness and AI urbanism [26], and today there is a debate regarding the relationship between these two visions of the urban [1, 52, 112]. Focusing our attention on AI, we can see how, on the one hand, this technology is referred to as a decisive tool for achieving sustainability. On the other hand, its very implementation and use are

interpreted as serious threats for a sustainable vision of human development. Admittedly, AI can be crucial to the efficiency of various types of systems and toward a reduction of squander and an optimization of energy trade-offs. From an urban point of view, for example, we can refer as much to waste as to water or electricity management, all the way to traffic flow management. Furthermore, the predictive capabilities of this technology can help city-managers to cope with extreme weather events and reduce industrial and agricultural emissions [18]. However, the link between AI and sustainability is not so peaceful. In fact, the material production of these technologies involves mining critical raw materials [24]; in addition, data processing and data storage are certainly not ecologically neutral [18]; finally recent studies are quantitatively emphasizing the considerable amount of emissions caused by the training of machine learning systems [105]. These elements make it difficult to conceive the connection between AI and sustainability in a linear and streamlined way [31]. An AI for sustainability that uses this technology to achieve sustainable goals needs to be coupled with a sustainable AI approach that can deal with emissions and ecological issues related to the entire life and production cycle of this technology [105].

Based on the above, it would be unwise to assume that an urbanism based primarily on increasing the capacity of UAI can in itself steer urban development toward sustainable goals. Recent efforts in the philosophy of technology and AI ethics to build frameworks that can hold AI and sustainability together witness a complicated nexus [18, 105]. The simple equation *more technology equals more sustainability* is fallacious. Indeed, a decisive theme in the literature on urban sustainability concerns the notion of balance. Steering urban development toward sustainability requires a balanced relationship between the technological component and the other components that inform the urban fabric (social, environmental, and economic issues) [52, 112]. The technocentric urban equation appears unbalanced and therefore ineffective when it comes to sustainability. Certainly, UAIs can be used to achieve economic, environmental and social, sustainability goals [112], however as we have seen, a purely technological approach is insufficient and AI can thus become an obstacle instead of a means to achieve sustainability [18, 105].

The transurbanist perspective is not capable of balancing the various urban elements. AI is conceived as the pre-dominant urban intelligence, capable of ordering and rationalizing all other components [11]. These components are not understood as intelligences, different from the artificial one, and capable of influencing its functioning and also thus giving direction to urban development. This last section of our charting process invites us to question and change the technocentric theoretical paradigm of AI implementation. This shift in perspective appears necessary to bring AI urbanism and sustainable urbanism together in the near future. A more promising theoretical framework seems to be that of the city understood as an ecology of intelligences, that is a paradigm supposed to interlink the various forms of urban agency and study their articulated relationships instead of attempting to reduce them all to the computational language of AI systems [69]. In this way, the application of UAIs must be calibrated against the needs of a specific set of urban agents and not deemed beneficial and necessary in every case. This perhaps more crafted and empirical approach capable of evaluating the pros and cons of UAI on a case-by-case basis, will only be made possible by questioning the technocentric paradigm. A sustainable approach to urban development in the digital age requires an ecological approach to the problem of urban intelligences, which can then guide the future of urban design.

2 Conclusion

The process of mapping AI urbanism across space and time lets us glimpse an articulated geography of relationships between the relevant points on the map. First, the project of a so-called science of the city is not a novel product of contemporary discourses but rather a constant theme, at least in Western modernity. The familiarity with this history that we have gained should make us reflect on the insufficiency of quantitative approaches for the representation of the city and thus for any action on it. Pondering the qualitative modes related to the data that are available today is one of the tasks that this map delivers to us. Moreover, we observed how this science of the city has long cultivated a dream that seemed to come true with the advent of cybernetics: an autonomous, objective and impartial functioning of urban technologies. However, we noted how such claimed autonomy risks becoming an automation of inequalities through bias and a misunderstanding of urban reality due to an underestimation of its contingent and spontaneous elements. Finally, the temporal axis of our reflection highlighted the risks of uncritically grafting the discourse of AI urbanism into a model such as the smart city paradigm, which has highlighted significant problems regarding the relationship between the interests of technology companies and the actual needs of urban realities.

Thereafter, through the spatial axis of our mapping, we understood both the structural relationship between the effective functioning of AI and the reorganization of space (enveloping) and brought to light the various material embodiments of urban AIs that we begin to relate to in our daily experience. This focus on spatiality has led us to ponder the design of UAIs not only from a technological point of view, by suggesting a more articulated and broader vision. Finally, this paper has challenged the idea that AI can be the dominant intelligence for the future of urbanity and that therefore a desirable urban development can be achieved merely by maximizing its use. UAI can be one of the forms of urban agency and an effective communicative partner for both citizens and planners if it is put in relation to the other intelligences (human and natural) scattered throughout our cities. Patiently building these connections, based on concrete needs seems to be a promising and more sustainable alternative to just increasing the quantity and quality of AI technology in the urban fabric. Thus, the connection between AI and urbanism emerges not as a simple implication but as an articulated relationship involving AI technology development, philosophy of technology, and urban studies. This interdisciplinarity emerges as necessary from the chart we have outlined. The charting proposed in this paper is an attempt to establish an example of a common ground of interdisciplinary reflection on the relationship between technology and the city in the age of AI. This relationship requires the various disciplines to step outside their epistemology in order to achieve a concrete and holistic capacity for analysis.

Certainly AI ethics, a distinctly philosophical discipline, could incorporate many of UAI's materiality insights in order to overcome some of its critical issues. In fact, after a major propulsive momentum, this discipline is now struggling to move beyond the stage of structuring general principles [39, 80] that can foster the ethical use of AI in practice. These general principles characterized by a certain abstractness can be easily circumvented by practitioners and politicians and are de facto often useless for effective regulation [83, 88]. The investigation of the material relationships in which UAIs are embedded could be useful in achieving a greater concreteness of the analysis concerning the ethics of AI at least in the urban context, by recognizing the social-technological and material dimensions of the relationships abstractly described in much of the current AI ethics literature. At the same time, ethical reflection on the implications of employing forms of artificial agency could help urban studies to consider not only the technical functionality but also the social and environmental contexts of UAI deployment. In fact, as we have seen, urban planning literature often focuses on the technological possibilities offered by these new systems while underestimating how their functioning can relate as much to communities as to the natural environment. In this regard, studies on the possibilities of human-machine relationships [17, 32, 40, 41] and AI sustainability [18, 105, 106] produced by the philosophy of technology could be fundamental to integrate UAI development into a sustainable urban design. Last but not least, STS could act as a bridging discipline between these two domains. Indeed, this strand of research shares with ethics a philosophical approach and a focus on the theoretical assumptions and implications that inform everyday practices, and with urban studies a marked focus on empirical research and concrete explorations of actually existing geographies, such as the relationships among the many intelligences situated in cities [23, 98, 100]. This barycentric epistemological position could be pivotal in balancing the necessary interdisciplinarity that the study of AI urbanism requires and that is proposed in this paper.

In conclusion, like any chart the one shaped in these pages is a tool in the hands of those who will use it. Such tool allowed us to interrogate the relationship between AI and urbanism by raising a number of crucial questions for the future of its development. However, like all maps, this one does not exhaust the object it represents and the paths indicated here can be followed, redeveloped or even challenged to further advance our understanding of AI urbanism. Situating the relationship between AI and urbanism within various cultures in order to understand their potentials and risks is an urgent task today if our goal is to transform mere technological development into a really sustainable urban innovation.

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Declarations

Competing interests The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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References

1. Ahvenniemi H, et al. What are the differences between sustainable and smart cities? *City*. 2016;60:234–45. <https://doi.org/10.1016/j.cities.2016.09.009>.
2. Albino V, Berardi U, Dangelico RM. Smart cities: definitions, dimensions, performance, and initiatives. *J Urban Technol*. 2015;22(1):3–21. <https://doi.org/10.1080/10630732.2014.942092>.
3. Allam Z. *The Rise of Autonomous Cities. Technology, Economic Performance and Climate Resilience*, Switzerland: Palgrave Macmillan; 2021.
4. Barns S. Out of the loop? On the radical and the routine in the urban big data. *Urban Stud*. 2021;58(15):3203–10. <https://doi.org/10.1177/00420980211014026>.
5. Bass T, Sutherland E, Symons T. Reclaiming the smart city. Personal data, trust and the new commons. Decode report. 2018. <https://decodeproject.eu/publications/reclaiming-smart-city-personaldata-trust-and-new-commons>.
6. Batty M. Artificial intelligence and smart cities. *Environ Plan B Urban Anal City Sci*. 2018;45(1):3–6. <https://doi.org/10.1177/2399808317751169>.
7. Bauman Z. *Liquid modernity*. Oxford: John Wiley & Sons; 2013.
8. Benevolo L. *Le origini dell'urbanistica moderna*. Bari: Laterza; 2022.
9. Birkinbine B. Commons praxis: towards a critical political economy of the digital commons. *TripleC*. 2018;1:290–305. <https://doi.org/10.31269/triplec.v16i1.929>.
10. Bostrom N. A history of transhumanist thought. *J Evol Technol*. 2005;14(1):1–25.
11. Bratton B. AI Urbanism: a design framework for governance, program and platform cognition. *AI & Soc*. 2021;36(4):1037–312.
12. Bria F. A new deal on data. In: McDonnell J, editor. *Economics for the many*. London: Verso; 2018. p. 164–73.
13. Bria F, Morozov E. *Ripensare la smart city*, Trad, it. Flavio Iannelli, Torino: Codice Edizioni; 2018.
14. Brovarone E, Scudellari J, Staricco L. Planning the transition to autonomous driving: a policy pathway towards urban liveability. *Cities*. 2021. <https://doi.org/10.1016/j.cities.2020.102996>.
15. Cerdà I. *Teoria generale dell'urbanizzazione*. Milano: Jaca Books; 1984.
16. Clever S, Crago T, Polka A, Al-Jaroodi J, Mohamed N. Ethical analyses of smart city applications. *Urban Sci*. 2018;2:96. <https://doi.org/10.3390/urbansci2040096>.
17. Coeckelbergh M. *AI Ethics*. Cambridge-London: MIT Press; 2020.
18. Coeckelbergh M. AI for Climate: freedom, justice and other ethical and political challenges. *AI Ethics*. 2021;1:67–72.
19. Coeckelbergh M. *The Political Philosophy of AI*, Cambridge: Polity, 2022.
20. Cohen T, Stilgoe J, Stares S, Akyelken N, Cavoli C, Day J, Dickinson J, Fors V, Hopkins D, Lyons G, Marres N, Newman J, Reardon L, Sipe N, Tennant C, Wadud Z, Wigley E. A constructive role for social science in the development of automated vehicles. *Transp Res Interdiscip Perspect*. 2020. <https://doi.org/10.1016/j.trip.2020.100133>.
21. Coletta C, Evans L, Heaphy L, Kitchin R (Eds.) *Creating Smart Cities*. London: London; 2018.
22. Collective C. Environments (Out) of control—notes on architecture's cybernetics entanglements. *Footprint*. 2021;15(1):81–99. <https://doi.org/10.7480/footprint.15.1.4942>.
23. Coutard O, Guy S. STS and the City. Policies and practices of hope. *Sci Technol Human Value*. 2007;32(6):716–34. <https://doi.org/10.1177/0162243907303600>.
24. Crawford K. *Atlas of AI. Power, politics and the planetary costs of artificial intelligence*. New Haven: Yale University Press; 2021.
25. Cugurullo F. Urban artificial intelligence: from automation to autonomy in the smart city, front. *Sustain Cities*. 2020;2:38. <https://doi.org/10.3389/frsc.2020.00038>.
26. Cugurullo F. *Frankenstein Urbanism. Eco, smart and autonomous cities, artificial intelligence and the end of the city*. New York: Routledge; 2021.
27. Cugurullo F. Good and evil in the autonomous city. In: Mackinnon D, Burns R, Fast V, editors. *Digital (in) justice in the Smart City*. University of Toronto Press; 2023. p. 183–94.
28. Currie M, Knox J, McGregor C. *Data justice and the right to the city*. GB: Edinburgh University Press; 2022.
29. Dooling R, McGuirk P. Autonomous vehicle experiments and the city. *Urban Geogr*. 2020. <https://doi.org/10.1080/02723638.2020.1866392>.
30. Duarte F, Álvarez R. The data politics of the urban age. *Palgrave Commun*. 2019;5:54. <https://doi.org/10.1057/s41599-019-0264-3>.
31. Dauvergne P. Is artificial intelligence greening global supply chains? Exposing the political economy of environmental costs. *Rev Int Political Econ*. 2022;29(3):696–718.
32. Esposito E. *Artificial Communication. How algorithms produce social intelligence*. Cambridge: MIT Press; 2022.
33. Fariás I, Bender T, editors. *Urban assemblages: How actor-network theory changes urban studies*. London: Routledge; 2009.
34. Fariás I. The politics of urban assemblage. *City*. 2011;15(3–4):365–74. <https://doi.org/10.1080/13604813.2011.595110>.
35. Fabbris L. Verso un'ecologia generale—Per una cibernetica delle differenze. *Philosophy Kitchen*. 2021;15:37–50. <https://doi.org/10.13135/2385-1945/621435>.
36. Ferraris M. *Documanità. Filosofia del nuovo mondo*. Bari: Laterza; 2021.

37. Floridi L. The fourth revolution: how the infosphere is reshaping human reality. UK: Oxford University Press; 2014. p. 37.
38. Floridi L, Cabitza F. L'uso umano delle nuove macchine. Milano: Bompiani; 2021.
39. Floridi L, Cows J. A unified framework of five principles for AI in society. *Harvard Data Sci Rev*. 2019. <https://doi.org/10.1162/99608f92.8cd550d139>.
40. Floridi L. Etica dell'intelligenza artificiale. Sviluppo, opportunità, sfide. Trad. It. Massimo Durante, Milano: Raffaello Cortina; 2022.
41. Floridi L. What the near future of artificial intelligence could be. *Philos Technol*. 2019;32:1–15. <https://doi.org/10.1007/s13347-019-00345-y41>.
42. Forrester JW. Urban dynamics. Cambridge: MIT Press; 1969.
43. Galdon-Clavell G. (Not so) smart cities? The drivers, impact and risks of surveillance-enabled smart environments. *Sci Public Policy*. 2013;40(6):717–23. <https://doi.org/10.1093/scipol/sct070>.
44. Greenfield A. Against the smart city: the City is Here for You to Use. New York: Verso; 2013.
45. Greenfield A. Radical technologies: the design of everyday life. London: Verso; 2017.
46. Golubchikov O, Thornbush M. Artificial intelligence and robotics in smart city strategies and planned smart development. *Smart Cities*. 2020;3(4):1133–44. <https://doi.org/10.3390/smartcities3040056>.
47. Graham S, Marvin S. Splintering urbanism: networked infrastructures, technological mobilities and the urban condition. London: Routledge; 2002.
48. Grigenti F. Le macchine e il pensiero. Napoli-Salerno: Orthotes; 2021.
49. Grossman RL, Heath A, Murphy M, Patterson M, Wells W. A case for data commons: toward data science as a service. *Comput Sci Eng*. 2016;18(5):10–20. <https://doi.org/10.1109/mcse.2016.92>.
50. Hall P. Cities of tomorrow. Oxford: Blackwell Publishers; 2002.
51. Harrison, et al. Foundations for smarter cities. *IBM J Res Dev*. 2010;54(4):1–16. <https://doi.org/10.1147/JRD.2010.2048257>.
52. Haupt W. The sustainable and the smart city: distinguishing two contemporary urban visions. In: Braers, R. Editor: The Palgrave Encyclopedia of Urban and Regional Futures; 2021.
53. Heeks R, Renken J. Data justice for development: what would it mean? *Inf Dev*. 2018;34(1):90–102. <https://doi.org/10.1177/0266666916678282>.
54. Heims SJ. The cybernetics group. London: MIT Press; 1991.
55. Hnilica S. The metaphor of the city as a thinking machine. A complicated relationship and its backstory. In S.M. Figueiredo, S. Krishnamurty e T. Schroeder Editors. *Architecture and the Smart City*, London: Routledge; 2020. pp. 68–85.
56. Hong Kong Government. Hong Kong Smart City Blueprint. 2018. <https://www.smartcity.gov.hk>.
57. Hörl E. A Thousand Ecologies: The Process of Cyberneticization and General Ecology. In D. Diederichsen & A. Franke Editors, *The Whole Earth: California and the Disappearance of the Outside*; Berlin: Sternberg Press; 2013, pp. 121–30.
58. IBM, Smarter cities, Smarter More Competitive Cities, Forward-thinking are investing in insight today, Somers, NY: IBM; 2012.
59. Jackman A, Brickell K. 'Everyday droning': towards a feminist geopolitics of the drone-home. *Prog Hum Geogr*. 2022;46(1):156–78. <https://doi.org/10.1177/03091325211018745>.
60. Jonas H. La cibernetica e lo scopo: una critica. Pisa: ETS; 1999.
61. Kassens-Noor E, Hintze A. Cities of the future? The potential impact of Artificial intelligence. *AI*. 2021;1(2):192–7. <https://doi.org/10.3390/ai1020012>.
62. Kitchin R. The real-time city? Big Data and smart urbanism. *GeoJournal*. 2014;79:1–14. <https://doi.org/10.1007/s10708-013-9516-8>.
63. Karvonen A, Cvetkovic V, Herman P, et al. The 'New Urban Science': towards the interdisciplinary and transdisciplinary pursuit of sustainable transformations. *Urban Transform*. 2021;3:9. <https://doi.org/10.1186/s42854-021-00028-y>.
64. Korinek A, Stiglitz J. Artificial Intelligence, Globalization, and Strategies for Economic Development", National Bureau of Economic Research, Working Paper, Working Paper Series, 28453, 2021
65. Korinek, Anton and Joseph E. Stiglitz, Artificial Intelligence and Its Implications for Income Distribution and Unemployment," In: Agrawal et al Editors. *The Economics of Artificial Intelligence*, NBER and University of Chicago Press; 2019.
66. Lee KF. AI superpowers. China, Silicon Valley and the New World Order, Boston: Houghton Mifflin Harcourt; 2018.
67. Lefebvre H. Writings on Cities: Henri Lefebvre. Selected, translated and introduced by E. Kofman and E. Lebas. Oxford: Blackwell; 1996.
68. Lobo J, Alberti M, Allen-Dumas M, Arcaute E, Barthelemy M, Bojorquez Tapia LA, et al. Urban science: integrated theory from the first cities to sustainable metropolises. Report submitted to the NSF on the present state and future of urban science; 2020. available at <https://ssrn.com/abstract=3526940>.
69. Lynch CR, Del Casino Jr VJ. Smart spacing, information processing, and the question of intelligence. *Ann Am Assoc Geogr*. 2020;110(2):382–90.
70. Maldonado T. Critica della ragione informatica. Milano: Feltrinelli; 1997.
71. Mattern S. City is not a computer: other urban intelligences. Princeton: Princeton University Press; 2021.
72. Mattern S. Code and clay, data and dust. Five thousand years of urban media. Minneapolis: Minnesota University Press; 2017.
73. McFarlane C, Söderström O. On alternative smart cities: from a technology intensive to a knowledge-intensive smart urbanism. *City*. 2017;21(3–4):312–28.
74. McCarroll C, Cugurullo F. Social implications of autonomous vehicles: a focus on time. *AI & Soc*. 2022;37:791–800. <https://doi.org/10.1007/s00146-021-01334-6>.
75. Mellas Z. Critical technics in architecture: a cybernetic approach. *Footprint*. 2021;15(1):99–111. <https://doi.org/10.7480/footprint.15.1.4936>.
76. Mezzadra S, Neilson B. On the multiple frontiers of extraction: excavating contemporary capitalism. *Cult Stud*. 2017;3(2–3):185–204. <https://doi.org/10.1080/09502386.2017.1303425>.
77. Mintrom M, et al. Robots in public space: implication for policy design. *Policy Design Practice*. 2022;5(2):123–39. <https://doi.org/10.1080/25741292.2021.1905342>.
78. Mitchell WJ. City of Bits. Space, Place, and the Infoban. Cambridge: MIT Press; 1996.

79. Mora L, Bolici R. *Progettare la smart city. Dalla ricerca teorica alla dimensione pratica*, Santarcangelo di Romagna: Maggioli Editore, 2016.
80. Morley J, Floridi L, Kinsey L, et al. From what to how: an initial review of publicly available AI ethics tools, methods and research to translate principles into practices. *Sci Eng Ethics*. 2020;26:2141–68. <https://doi.org/10.1007/s11948-019-00165-5>.
81. Mumford L. *Technics and Civilization*. New York: Harcourt Brace and Company; 1934.
82. Mumford L. *The city in history: its origins its transformations and its prospects*. San Diego: Harcourt Brace; 1961.
83. Munn L. The uselessness of AI ethics. *AI Ethics*. 2022. <https://doi.org/10.1007/s43681-022-00209-w>.
84. Pascal A. The vanishing city. *Urban Stud*. 1987;24(6):597–603. <https://doi.org/10.1080/00420988720080821>.
85. Pawley M. *Architecture, urbanism and the new media*. New York: Mimeo; 1995.
86. Perng S, Maalsen S. Civic infrastructure and the appropriation of the corporate smart city. *Ann Am Assoc Geogr*. 2020;110(2):507–15. <https://doi.org/10.1080/24694452.2019.1674629>.
87. Picon A, Hill T. Is the city becoming computable? In *Architecture and the Smart City*. In: Figueiredo SM, Krishnamurty S, Schroeder T, editors. *Architecture and the Smart City*. London: Routledge; 2020. p. 29–43.
88. Rességuier A, Rodrigues R. AI ethics should not remain toothless! A call to bring back the teeth of ethics. *Big Data Soc*. 2020. <https://doi.org/10.1177/2053951720942541>.
89. Robbins S. AI and the path to envelopment: knowledge as a first step towards the responsible regulation and use of AI-powered machines. *AI & Soc*. 2020;35:391–400. <https://doi.org/10.1007/s00146-019-00891-1>.
90. Rosen E, Garboden PM, Cossyleon JE. Racial discrimination in housing: how landlords use algorithms and home visits to screen tenants. *Am Sociol Rev*. 2021;86(5):787–822. <https://doi.org/10.1177/00031224211029618>.
91. Sennett R. *Building and dwelling: ethics for the city*. London: Allen Lane an imprint of Penguin Books; 2018.
92. Shapiro A. Predictive policing for reform? Indeterminacy and intervention in big data policing. *Surveill Soc*. 2019. <https://doi.org/10.24908/ss.v17i3/4.10410>.
93. Shelton T, Zook M, Wiig A. The ‘actually existing smart city’. *Camb J Reg Econ Soc*. 2015;8(1):13–25. <https://doi.org/10.1093/cjres/rsu026>.
94. Scholte T, Sweeting B. Possibilities for a critical cybernetics. *Syst Res Behav Sci*. 2022. <https://doi.org/10.1002/sres.2891>.
95. Sejnowski JT. *The deep learning revolution*. Cambridge: MIT Press; 2018.
96. Söderström O, Mermet A-C. When airbnb sits in control room: platform urbanism as actually existing smart urbanism in Reykjavik. *Front Sustain Cities*. 2020;2:15. <https://doi.org/10.3389/frsc.2020.00015>.
97. Söderström O, Paasche T, Klauser F. Smart city as “Corporate Storytelling”. *City*. 2014;18:317–20.
98. Stilgoe J. Machine learning, social learning and the governance of self-driving cars. *Soc Stud Sci*. 2018;48(1):25–56. <https://doi.org/10.1177/0306312717741687>.
99. Sumartojo S, Lundberg R, Kulić D, Tian L, Carreno-Medrano P, Mintrom M, et al. The robotic production of spatiality: predictability, partitioning, and connection. *Trans Inst Br Geogr*. 2022.
100. Tennant C, Stilgoe J. The attachments of ‘autonomous’ vehicles. *Soc Stud Sci*. 2021;51(6):846–70. <https://doi.org/10.1177/03063127211038752>.
101. Tiddi I, Bastianelli E, Daga E, et al. Robot-city interaction: mapping the research landscape—a survey of the interactions between robots and modern cities. *Int J Soc Robot*. 2020;12:299–324. <https://doi.org/10.1007/s12369-019-00534-x>.
102. Trencher G. Towards the smart city 2.0: empirical evidence of using smartness as a tool for tackling social challenges. *Technol Forecast Soc Change*. 2019;142:117–28. <https://doi.org/10.1016/j.techfore.2018.07.033>.
103. Tulumello S, Iapaolo F. Policing the future, disruptive urban policy today. Predictive policing, smart city, and urban policy in Memphis (TN). *Urban Geogr*. 2021;43(3):448–69. <https://doi.org/10.1080/02723638.2021.1887634>.
104. Vanolo A. Is there anybody out there? The place and role of citizens in tomorrow’s smart cities. *Futures*. 2016;82:26–36. <https://doi.org/10.1016/j.futures.2016.05.010>.
105. Van Wynsberghe A. Sustainable AI: AI for sustainability and the sustainability of AI, in *AI and Ethics*. *Sustainability*. 2021;1:213–8. <https://doi.org/10.1007/s43681-021-00043-6>.
106. Van Wynsberghe A, Vandemeulebroucke T, Bolte L, Nachid J. Special issues “Towards the Sustainability of AI; Multi-Disciplinary Approaches to Investigate the Hidden Costs of AI.” *Sustainability*. 2022;14(24):16352. <https://doi.org/10.3390/su142416352>.
107. Verdegem P. Dismantling AI capitalism: the commons as an alternative to the power concentration of Big Tech. *AI & Soc*. 2022. <https://doi.org/10.1007/s00146-022-01437-8>.
108. While AH, Marvin S, Kovacic M. Urban robotic experimentation: San Francisco, Tokyo and Dubai. *Urban Stud*. 2021;58(4):769–86. <https://doi.org/10.1177/0042098020917790>.
109. Wiener N. *Cybernetics, or control and communication in the animal and the machine*. Cambridge: MIT Press; 1965.
110. Willis K, Aurigi A. *Digital and smart cities*. London: Routledge; 2017.
111. Winner, L. *Autonomous Technology. Technics-out-of-control as a Theme in Political Thought*, US: MIT Press, 1977.
112. Yigitcanlar T, Cugurullo F. The Sustainability of Artificial intelligence: an Urbanistic Viewpoint from the Lens of Smart and Sustainable Cities. *Sustainability*. 2020;12:20. <https://doi.org/10.3390/su12208548>.
113. Zaffagnini T, Palmini O. Past and future of the connection between project, technology and neocybernetics. *J Agathòn*. 2022. <https://doi.org/10.19229/2464-9309/1222022>.
114. Zuboff S, Schwandt K. *The age of surveillance capitalism: the fight for a human future at the new frontier of power*. London: Profile Books; 2019.
115. Zucconi G. *La città dell’Ottocento*. Bari: Laterza; 2022.